

Lesson 1.1 — What “AI-Assisted Development” Actually Means in 2026

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If your interviewer asks “How do you use AI in your work?” and you say “I use Copilot for autocomplete,” you’ve already lost the room. This lesson teaches you to answer that question like a senior engineer.

By the end of this lesson:

- You’ll have a precise, vocabulary-rich definition of AI-assisted development.
- You’ll know the difference between **autocomplete**, **agentic**, and **conversational** AI workflows — and which one your interviewer probably uses.
- You’ll know how to use the phrase “vibe coding” without getting yourself rejected.
- You’ll have a 30-second self-introduction line you can use in any AI-context interview.

1. The five generations of AI tooling (and where you sit)

To talk about AI-assisted development at the senior level, you need to know that there have been roughly five overlapping generations of tools. Most engineers conflate them. Most interviewers don’t. Knowing the difference is a marker of seniority.

Generation	Era	Example tools	What it gives you
G1: Autocomplete	2021–2022	GitHub Copilot v1, TabNine	Single-line and multi-line completions inside the editor

Generation	Era	Example tools	What it gives you
G2: Chat	2023	ChatGPT, Claude.ai, Gemini	A separate window where you ask questions and paste code
G3: In-IDE Chat	2024	Copilot Chat, Cursor, Continue	A side panel that knows about your file and selection
G4: Agentic IDE	2024–2025	Cursor Composer, Windsurf, Aider, Cline	A loop that <i>reads, edits, runs commands</i> , and <i>iterates</i> across multiple files
G5: Autonomous agent	2025–2026	Devin, OpenAI Operator, Cascade in plan mode	Long-horizon tasks: “implement this feature, run tests, open a PR”

Most interviewers in 2026 expect you to be fluent in G3 and G4. If you can articulate the difference between agentic and chat-based workflows, you’re in the top 25% of candidates already.

How to drop this casually in an interview

“I’m primarily on G4 – agentic editing. I use it for multi-file refactors and data-shaping. For one-off questions or unfamiliar APIs I’ll drop into a G2/G3 chat. I haven’t moved fully to G5 autonomous agents yet because I want to review every diff before it lands.”

That sentence does five things at once: it shows you know the taxonomy, names the tools generically (no vendor-shilling), demonstrates judgment about which generation is right for which task, and signals that you take code review seriously. **Use it.**

2. The three workflows you should be able to name

Underneath those five tool generations are three actual *workflows*. Senior engineers know the workflows; junior engineers know only the tools.

2.1. Generation workflow

You know what you want. You ask. The AI produces it. You review and accept.

Example: *“Generate a TypeScript interface for a user with id, email, name, and createdAt as ISO 8601 string.”*

- **You own:** the spec, the review, the commit.
- **AI owns:** typing the lines.
- **Skills required:** precise specification, code review, type discipline.

This is the workflow most people **think** AI-assisted development is. It’s actually the simplest one and the one with the lowest leverage.

2.2. Exploration workflow

You don’t know what you want. You explore options with the AI as a thinking partner.

Example: *“I have a list of products with categories. I want a UI that lets users filter and search. What are three patterns for this and what are the trade-offs?”*

- **You own:** the question, the judgment about which pattern fits your codebase, the final pick.
- **AI owns:** surfacing options you might not have considered, summarizing trade-offs.
- **Skills required:** curiosity, the ability to evaluate trade-offs, and the discipline to pick *one* and commit.

This is the highest-leverage workflow most engineers underuse. Senior engineers use it heavily before any code gets written.

2.3. Iteration workflow

The AI generated something. It’s 80% there. You and the AI iterate until it’s right.

Example: *“This works but the loading state flickers when the data is cached. Fix it without changing the public API.”*

- **You own:** the failure observation, the desired behavior, the regression check.
- **AI owns:** the diff.
- **Skills required:** debugging instincts, the ability to specify a fix without specifying *the* implementation, and patience.

This is the workflow that **actually fills your day**. It's also the workflow interviewers most want to test.

How to drop this in an interview

"I think of AI-assisted work in three modes: generation when I know exactly what I want, exploration when I'm trying to choose between approaches, and iteration when I'm refining something that already exists. Most of my real work is iteration."

That's a quote-grade answer. Memorize the structure.

3. The "vibe coding" minefield

Andrej Karpathy coined the phrase "vibe coding" in early 2025: *"I 'vibe code' – I just see stuff, say stuff, run stuff, and copy-paste stuff, and it mostly works."*

Within months it became:

- A celebration on Twitter.
- A pejorative in code review.
- A loaded term in interviews.

You will get asked about vibe coding. Here is how to handle it.

3.1. What "vibe coding" means at the *good* end

- Iterating quickly with AI assistance.
- Trusting your taste about whether the output looks right.
- Skipping ceremony when ceremony doesn't earn its keep.
- Letting the LLM type so you can think.

3.2. What "vibe coding" means at the *bad* end

- Shipping code you can't read.
- "It works" being the only test.
- Not understanding the abstractions the AI introduced.
- Copying solutions to problems you couldn't articulate.

3.3. How interviewers use the phrase

Interviewers asking “Are you a vibe coder?” are usually asking one of three different questions:

1. “Are you the bad kind of vibe coder?” ☒ They want to confirm you read what the AI writes.
2. “Are you the good kind?” ☒ They want to confirm you can move fast with AI without ceremony paralysis.
3. “Do you know what the term means?” ☒ They’re testing if you’ve kept up with the field.

The right answer is one that disambiguates. Try this:

“There’s a good and a bad version of that phrase. The bad version is shipping code you can’t explain. The good version is iterating fast with AI as your typist while you stay the engineer. I aim for the good version. Concretely: I always read the diff before accepting, I write tests for AI-generated logic before I trust it, and if I can’t explain a function the AI gave me, that’s a flag — I don’t accept it until I can.”

That answer:

- Acknowledges the phrase exists.
- Distinguishes the two interpretations.
- Stakes a clear position.
- Backs the position with three concrete behaviors.

It will satisfy almost any interviewer who asks. **Drill this answer.**

4. The three categories of work AI is genuinely good at

You’ll be asked: “What do you trust AI to do?” Have a structured answer.

4.1. Pattern-rich, low-novelty work

- Boilerplate (form scaffolds, CRUD endpoints, type definitions).
- Conversions (JSON ☒ TS interfaces, CSS ☒ Tailwind, Python ☒ Go).
- Test scaffolding (writing the *next* test that follows the existing pattern).
- Documentation (turning code into prose, generating JSDoc/docstrings).

AI is very good at these. They're high-volume in real work. **Trust them, review them, ship them.**

4.2. Constrained, well-specified algorithms

- Implementing a known algorithm given a clear spec ("debounce", "trie", "merge sort").
- Glue code between well-known APIs (file → S3, webhook → handler).
- Refactors with a clear target ("extract this into a custom hook").

AI is good at these *if you specify them well*. **The quality of the output is a function of the quality of the prompt.**

4.3. Search and synthesis

- "What are the three ways people typically solve X?"
- "What does this error message usually mean in Postgres?"
- "Compare TanStack Query vs SWR for my use case."

AI is faster than Stack Overflow and more current than your memory. **It's a research tool now**, even when it doesn't write code.

5. The three categories AI is genuinely bad at (still)

This is the **other** half of the answer interviewers want. If you only list strengths, you sound naive. If you list weaknesses with examples, you sound senior.

5.1. Novel architecture under real constraints

The LLM doesn't know your team's deploy story, your billing tier, your historical war stories about the queue that took down prod, your manager's risk tolerance, your timeline. It will produce a design that looks reasonable in isolation and is wrong for *your* context.

5.2. Cross-cutting refactors that span a long history

Imagine: "rename a field across 60 files, but only in the new code paths, leaving the legacy paths alone." The LLM doesn't have the institutional memory of which paths are legacy. It

will rewrite the wrong subset, confidently.

5.3. Debugging where the bug is in your *understanding*

If you're prompting the AI to "fix this race condition" but you've misdiagnosed which two operations are racing, the AI will fix the wrong race condition and the bug will return tomorrow.

6. Your 30-second self-intro line

Combine the above into one paragraph you can deliver in any interview that opens with "Tell me about your relationship with AI tools":

"I've been on agentic AI tooling — currently [name a tool you actually use] — for about [N] months. I use it across three modes: generation, exploration, and iteration. Most of my day is iteration. The work I trust AI most for is pattern-rich code — types, tests, conversions, scaffolding. The work I keep close is novel architecture under team-specific constraints, anything that requires reading a long history of the codebase, and debugging where I might have the wrong mental model. I always read the diff before accepting it, and if I can't explain a function the AI wrote, I don't ship it."

That's ~75 words, ~30 seconds. It hits:

- Tools (without vendor-shilling).
- Workflows (named, not vague).
- Strengths (concrete examples).
- Limits (concrete examples).
- A one-line ethical commitment ("I read the diff").

Memorize this. It is your opening.

7. Three pitfalls when answering AI questions

7.1. The “I love AI” pitfall

You sound like a fan, not an engineer. The interviewer will dig deeper to see if you have any judgment.

7.2. The “I’m anti-AI” pitfall

You sound like you’re hiding the fact that you don’t use it. The interviewer will assume you’ll be slower than the AI-fluent candidate.

7.3. The “It depends” pitfall

You sound evasive. “**It depends**” is only a great answer when you immediately follow with the things it depends on.

Bad: “Do you trust AI for production code?” ☒ “It depends.”

Good: “Do you trust AI for production code?” ☒ “It depends on three things: how high the blast radius is, how good my test coverage is around the change, and how well the AI seemed to understand the existing patterns. For a CRUD endpoint with 90% test coverage, yes. For the billing reconciliation job? I’d review every line and write the tests myself.”

8. CHECK YOURSELF

Don’t move on until you can answer these out loud:

- ☐ What are the five generations of AI tooling? Which one are *you* primarily on?
- ☐ What are the three workflows? Which one is most of your real work?
- ☐ What are the three things AI is genuinely good at? Give one concrete example for each.
- ☐ What are the three things AI is genuinely bad at? Give one concrete example for each.
- ☐ How do you respond to “Are you a vibe coder?” without dismissing the term or sounding cocky?
- ☐ Can you deliver your 30-second self-intro line, twice in a row, without notes?

If any of those is fuzzy, scroll back. The next lesson assumes this vocabulary.

9. Where you are now

You have:

- A taxonomy of AI tools that puts you in the top quarter of candidates by vocabulary alone.
- A workflow model that lets you describe what you actually do all day.
- A defended answer to the most loaded phrase in the field (“vibe coding”).
- A 30-second opening that signals seniority in the first sentence.

Move on to [02-the-skill-stack.md](#) to map out which skills are now AI’s, which are still yours, and which are *new* skills you should be selling in interviews.